

# Ultra-high-speed dynamics of acoustic droplet vaporization in soft biomaterials: Effects of viscoelasticity, frequency, and bulk boiling point

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## ABSTRACT

Phase-shift droplets are a highly adaptable platform for biomedical applications of ultrasound. The spatiotemporal response of phase-shift droplets to focused ultrasound above a certain pressure threshold, termed acoustic droplet vaporization (ADV), is influenced by intrinsic features (e.g., bulk boiling point) and extrinsic factors (e.g., driving frequency and surrounding media). A deep understanding of ADV dynamics is critical to ensure the robustness and repeatability of an ADV-assisted application. Here, we integrated **ultra-high-speed imaging, at 10 million frames per second**, and confocal microscopy for a full-scale (i.e., from nanoseconds to seconds) characterization of ADV. Experiments were conducted in **fibrin-based hydrogels** to mimic **soft tissue environments**. Effects of **fibrin concentration (0.2 to 8 % (w/v))**, **excitation frequency (1, 2.5, and 9.4 MHz)**, and **perfluorocarbon core** (perfluoropentane, perfluorohexane, and perfluorooctane) on ADV dynamics were studied. Several fundamental parameters related to ADV dynamics, such as **expansion ratio, expansion velocity, collapse radius, collapse time, radius of secondary rebound, resting radius, and equilibrium radius of the generated bubbles** were **extracted from the radius vs time curves**. Diffusion-driven ADV-bubble growth was fit to a **modified Epstein-Plesset equation**, adding a material stress term, to estimate the growth rate. Our results indicated that ADV dynamics were significantly impacted by fibrin concentration, frequency, and perfluorocarbon liquid core. This is the first study to combine ultra-high-speed and confocal microscopy techniques to provide insights into ADV bubble dynamics in tissue-mimicking hydrogels.

## 1. Introduction

Ultrasound-responsive droplets, termed phase-shift droplets, have advanced the field of biomedical ultrasound. Phase-shift droplets consist of a perfluorocarbon liquid core stabilized by a shell. Perfluorocarbons are fluorinated compounds widely used in the past as oxygen carriers due to their inertness and high oxygen solubility [1]. Phase-shift droplets possess distinct advantages compared with conventional contrast microbubbles such as improved stability and longer circulation half-life in vivo [2,3]. These liquid droplets maintain their liquid state at physiologic temperature due to a combination of increased Laplace pressure [4] and inhibition of heterogeneous nucleation [5]. An ultrasound pulse triggers activation and phase-transition into a gaseous state – a process termed acoustic droplet vaporization (ADV) [6]. Since ADV enables a highly localized and non-invasive generation of bubbles in situ in a

spatiotemporally controlled manner, it has been extensively researched in ultrasound-based diagnostic [7–9] and therapeutic applications [10–13].

The underlying complex physics and the critical parameters affecting ADV have been the subject of several studies. A deep understanding of the dynamic response of phase-shift droplets to ultrasound is critical to ensure robust and repeatable outcomes for any ADV-assisted application. Both experimental and theoretical approaches have been undertaken to study the phase-transition and ADV-generated bubble dynamics under varying ultrasound conditions. Although great progress has been made on developing mathematical models to predict various stages of ADV, from phase-transition [14–16] to bubble growth [17–19], real-time experimental assessment of ADV dynamics remain limited due to the transient and microscopic nature of the phenomenon. High-speed, brightfield microscopy has enabled study of ADV at high temporal

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resolutions. Formation of vapor nucleus in the initial phase-transition at the nanosecond timescale, interaction of the generated bubbles with ultrasound, and the growth stage microseconds post ultrasound have been recorded [20]. High-speed imaging, at a nanosecond timescale, demonstrated that the initial vapor nucleus was generated within the droplet, through homogeneous nucleation, and it was along the direction of ultrasound propagation [21–23]. Further studies showed that the initial nucleation site depended on the interplay between the size of the droplet and the wavelength of the incoming wave [22]. The connection between the initial nucleation site and the pressure and phase information of the incident ultrasound pulse indicated that the nucleation took place during the peak rarefactional half cycle of the ultrasound when the liquid experienced tension [23]. More recent high-speed studies have focused on the dynamics of the generated ADV-bubbles since their response to ultrasound is paramount to the biomedical applications. For example, the expansion ratio, defined as the maximum bubble radius with respect to the initial radius of the droplet, expansion velocity, and final bubble size correlated inversely with perfluorocarbon bulk boiling point [17]. Expansions as high as  $\sim 8$ -fold were reported for droplets with bulk boiling points less than 29 °C. Microseconds after ultrasound was turned off, the ADV-generated bubbles were shown to oscillate in a decaying manner, due to unforced (i.e., not driven by ultrasound) radial oscillation, and settled to a final resting bubble size [24,25]. A detailed review of high-speed microscopy studies of ADV dynamics is provided in a prior review article [20].

Prior research on ultra-fast dynamics of ADV has been exclusively conducted in Newtonian fluids. Yet, biological liquids and soft tissues exhibit non-Newtonian behavior [26], necessitating investigation of ADV dynamics and relevant parameters in a viscoelastic medium. Additionally, a full-scale characterization of ADV dynamics across a broad range of time scales, from nanoseconds to minutes, is required to fully capture the dynamics and mechanical interactions involved. Acoustic emissions during the microsecond ultrasound pulses, which are significantly affected by the radial dynamics of the ADV bubbles, play a critical role in diagnostic ultrasound. For applications such as tissue engineering, a comprehensive understanding of the longer-term behavior of the generated bubbles is also essential.

In this study, for the first time, we integrated ultra-high-speed microscopy and confocal microscopy to provide a full-scale characterization of ADV dynamics from nanoseconds to seconds timescales. We investigated the ADV dynamics of three common phase-shift droplets – perfluoropentane (PFP), perfluorohexane (PFH), and perfluorooctane (PFO) – in fibrin-based hydrogels. Effects of fibrin concentration, excitation frequency, and bulk boiling point of phase-shift droplets on ADV dynamics were investigated. Fibrin was chosen as the hydrogel for its abundance as a provisional matrix during wound healing, FDA approval as a hemostatic sealant, and its broad range of biomedical applications [27,28]. The fibrin concentration range was chosen to mimic soft tissues with elastic moduli lower than 10 kPa. This approach is crucial in enhancing our understanding of how the surrounding microenvironment impacts the ADV response. Such elucidation is pivotal for advancing the development of innovative *in situ* micro-rheology methods. These methods are particularly useful in the context of pathological conditions such as fibrosis and cancer, where local mechanical properties of tissues change overtime. PFP droplets require low acoustic energies for activation (i.e., low ADV threshold), and thus have

been used in ultrasound imaging applications [9]. Droplets with comparatively higher bulk boiling points are predominantly used in therapeutic applications such as drug delivery and tissue regeneration [10,11]. Therefore, the findings here provide useful insight for both therapeutic and diagnostic applications of ultrasound. For example, in drug delivery applications, on-demand modulation of critical parameters such as dose and release rate remains a significant challenge. Designing payload-carrying phase-shift droplets and tailoring their response to ultrasound to achieve desired release kinetics necessitates a thorough understanding of the underlying physics. Several fundamental parameters, such as expansion ratio, collapse time, collapse radius, and radius at quasistatic equilibrium, were derived from our ultra-high-speed and confocal studies. This paper is the first to provide an in-depth understanding of ADV dynamics by analyzing fundamental parameters extracted from radius vs time curves. These key parameters are important for optimizing the use of phase-shift droplets and tuning the mechanical effects in a variety of biomedical applications.

## 2. Materials and methods

### 2.1. Microfluidic production of phase-shift droplets

Monodisperse microdroplets were generated via a microfluidic-based chip (Cat# 3200146, Dolomite, Royston, United Kingdom). The details of production of phase-shift microdroplets can be found in our prior publications [29,30]. PFP (CAS# 355–42-0, Strem Chemicals), PFH (CAS# 355–42-0, Strem Chemicals) or PFO (CAS# 307–34–6, Sigma-Aldrich) was used as the perfluorocarbon phase. For ease of visualization of droplets during confocal imaging, Cascade Blue dextran (Thermo Fisher Scientific, Waltham) was used. A solution of 50 mg/mL Pluronic F68 (CAS# 9003–11-6, Sigma-Aldrich) in phosphate buffer saline (PBS, Thermo Fisher) encapsulated the droplets. To produce phase-shift droplets with a nominal diameter of  $\sim 12$   $\mu$ m, flow combination was set at 0.5  $\mu$ L/min and 2.5  $\mu$ L/min for the perfluorocarbon containing phase and the Pluronic F68 solution, respectively. Due to its relevance to our prior publications as well as other studies within the tissue regeneration community, we used larger size phase-shift droplets. In this work, we refer to each phase-shift droplet by its respective perfluorocarbon (e.g., PFP droplets). Size characteristics of the droplets, measured using a Coulter Counter (aperture tube: 30  $\mu$ m, Multisizer 4, Beckman Coulter), are summarized in Table 1.

### 2.2. Preparation of fibrin-based hydrogels

Details of preparing fibrin hydrogels containing phase-shift droplets are described in our prior publications [28,29]. Briefly, bovine fibrinogen solution (Sigma-Aldrich) was prepared at a concentration of 80 mg/mL in Dulbecco's modified Eagle's medium (DMEM, Thermo Fisher Scientific). The fibrinogen solution was then degassed for an hour. Fibrin-based hydrogels were prepared by combining the fibrinogen solution, DMEM, thrombin (Recothrom, Baxter, Deerfield, IL, USA), and phase-shift droplets. To polymerize fibrin hydrogels, thrombin with a concentration of 2 U/mL was added in the final step. We used a low volume fraction of droplets (0.001 % (v/v)) to minimize inter-droplet interactions (inter-droplet spacing:  $\sim 260$   $\mu$ m [34]). To study the effect of fibrin concentration on ADV dynamics of PFP droplets, hydrogels with

**Table 1**

Size characteristics of perfluorocarbon (PFC) phase-shift droplets used to study ultra-high-speed dynamics of acoustic droplet vaporization in fibrin-based hydrogels. Three PFC liquids were used: perfluoropentane (PFP), perfluorohexane (PFH) and perfluorooctane (PFO). Concentrations of droplets were  $(0.1\text{--}0.3) \times 10^9$  particles/mL. Thermophysical properties of PFC liquids are referenced from [31–33].

| PFC liquid | Chemical formula               | Bulk boiling point (°C) | Vapor pressure (kPa) | Radius ( $\mu$ m) | Coefficient of variation (%) |
|------------|--------------------------------|-------------------------|----------------------|-------------------|------------------------------|
| PFP        | C <sub>5</sub> F <sub>12</sub> | 29                      | 135                  | 6.2 $\pm$ 0.2     | 3.5                          |
| PFH        | C <sub>6</sub> F <sub>14</sub> | 56                      | 46                   | 6.3 $\pm$ 0.2     | 3.7                          |
| PFO        | C <sub>8</sub> F <sub>18</sub> | 100                     | 0.8                  | 6.7 $\pm$ 0.1     | 2.6                          |

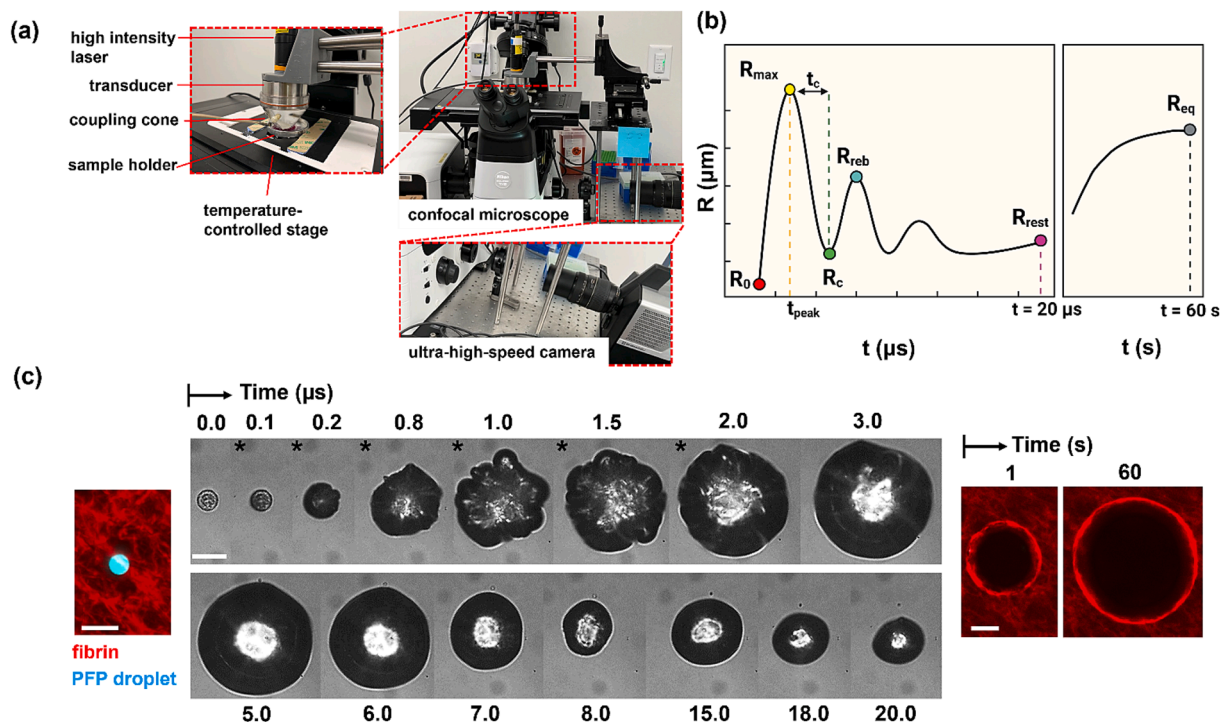
varying fibrin concentrations of 0.2 to 8 % (w/v) were prepared. Shear elastic and loss moduli of fibrin hydrogels were measured using an AR-G2 rheometer (TA instruments, New Castle, USA) with a parallel plate geometry (measurement head diameter: 8 mm, gap height: 2000  $\mu\text{m}$ , temperature: 37  $^{\circ}\text{C}$ ). Samples were then subjected to oscillatory shear at 1 Hz with a strain of 1 %, based on the linear viscoelastic regime determined previously [30], to obtain shear elastic and loss moduli. The concentrations were chosen to mimic soft tissue properties [35,36]. For example, elastic moduli of 0.3–1.6 kPa, 0.6–1.7 kPa, and 8.7–27.8 kPa have been reported for brain, liver, and muscle tissues, respectively [37,38]. Hydrogels with a fibrin concentration of 1 % (w/v) were prepared to investigate the effect of excitation frequency and perfluorocarbon bulk boiling point. Fibrin was fluorescently labeled with 39  $\mu\text{g}/\text{mL}$  Alexa Fluor 647-labeled fibrinogen (F35200, Molecular Probes, Eugene, OR, USA) for confocal imaging. Hydrogels (height:  $\sim 2$  mm) were directly polymerized in a custom-made, 3D-printed (Form 3L, Formlabs, Boston, USA) cylinder with an opening in the bottom for a cover slip. DMEM was used as the overlying media to couple with the ultrasound transducer.

### 2.3. Experimental setup and parameters

Focused single-element transducers (H-101 ( $f\#$ : 0.98), and H-147 ( $f\#$ : 0.83), Sonic Concepts Inc., Bothell, WA, USA) were used to induce ADV. Transducers were mounted on a 3-axis translation stage to confocally align the acoustic and optical foci. Calibration was performed in free field using an in-house built fiber optic hydrophone as described before [39,40]. Acoustic pressures are reported as peak rarefactional pressures in the manuscript ( $P_r$ ). Each transducer was driven by a single, sine-wave ultrasound burst generated by a waveform generator

(33220A, Agilent Technologies). Generated signals were amplified by a radiofrequency amplifier (240L, Electronic and Innovation Ltd). Experiments were conducted at three clinically relevant excitation frequencies: 1 MHz ( $P_r$ : 4 MPa), 2.5 MHz ( $P_r$ : 4 MPa), and 9.4 MHz ( $P_r$ : 4 MPa). A higher pressure was used to generate ADV in PFO droplets at 2.5 MHz ( $P_r$ : 4.5 MPa). The selected  $P_r$  was suprathreshold for ADV for all phase-shift droplets studied herein. In fluids such as water, inertial cavitation (IC) occurs when the generated bubble grows to at least twice its original diameter, followed by a violent collapse driven by the inertia of the surrounding media. We previously showed that ADV occurred at lower rarefactional pressures than IC, indicating that phase transition precedes IC [40]. However, at lower frequencies and in phase-shift droplets with higher bulk boiling points, ADV and IC could happen at similar acoustic pressures [41]. The selected  $P_r$  was above the IC threshold for phase-shift droplets at 1 MHz and 2.5 MHz [40,42]. In our prior studies IC thresholds of  $1.7 \pm 0.2$  MPa and  $5.6 \pm 0.4$  MPa were measured in fibrin-only hydrogels at 1.1 MHz (pulse repetition frequency (PRF): 100 Hz) and 2.5 MHz (PRF: 100 Hz), respectively [40,41]. However, in the current study no IC was observed in fibrin-only hydrogels due to the use of single burst ultrasound. The number of cycles was kept constant ( $N = 6$ ) throughout the experiments.

For ultrasound exposure, the mounted transducer was coupled with hydrogels containing phase-shift droplets through 3D-printed coupling cones filled with deionized water (Fig. 1a). An ultra-high-speed camera (HPV-X2, Shimadzu), providing a total of 256 frames at frame rates up to 10 million frames per second (Mfps), was paired with the confocal microscope (AX, Nikon) with a 20x objective (PLAN APO, N.A: 0.8, Nikon) (Fig. 1a). A high-intensity pulsed laser (400 W, 640 nm, Cavitar Ltd) was used in a transmittance configuration for high intensity, strobed illumination. The collimated illumination laser was aligned with the



**Fig 1.** Ultra-high-speed and confocal microscopy techniques were integrated with focused ultrasound to study the dynamics of acoustic droplet vaporization (ADV) under varying conditions in soft biomaterials composed of fibrin. ADV dynamics of three commonly used phase-shift droplets were studied: perfluoropentane (PFP), perfluorohexane and perfluorooctane. (a) Pictures of the integrated experimental setup showing the confocal microscope, ultra-high-speed camera, and focused ultrasound transducer. Ultra-high-speed imaging was performed in a back-illumination configuration at frame rates up to 10 million frames per second. Confocal microscopy was performed at 1 frame per second. (b) Images were processed to plot the temporal evolution of bubble radius, capturing bubble expansion, oscillation, collapse, and subsequent rebounds as shown schematically. Parameters such as initial radius ( $R_0$ ), maximum expansion radius ( $R_{\max}$ ), minimum radius ( $R_c$ ), secondary rebound radius ( $R_{\text{reb}}$ ), radius at the end of high-speed imaging (i.e., at 20  $\mu\text{s}$ ) ( $R_{\text{rest}}$ ), and radius at the quasistatic equilibrium (i.e., at 60 s) ( $R_{\text{eq}}$ ) were derived and compared for varying conditions. (c) Selected ultra-high-speed and confocal image sequences of a PFP droplet embedded in a 0.2 % (w/v) fibrin hydrogel exposed to a single burst of 2.4  $\mu\text{s}$  at 2.5 MHz ( $P_r$ : 4 MPa). Presence of ultrasound in the field of view is denoted with an asterisk. Scale bar: 25  $\mu\text{m}$ .

transducer with a 20-mm opening using a combination of two 50-mm plano-convex lenses, and two infinitely-corrected tube lenses (Thorlabs, New Jersey, USA). Exposure time was set to 50 ns. High-speed video frames were analyzed to determine the bubble radius at every time point (Fig. 1b & c) using a custom image processing algorithm developed in MATLAB (The MathWorks) and TEMA Pro (Image Systems, Linköping, Sweden), an advanced, commercially available motion tracker software. Based on the speed of sound in water at 37 °C and the focal length of the transducers, the camera recording was delayed appropriately to start a few nanoseconds before the arrival of ultrasound and acquired a total window of 25.6  $\mu$ s. The synchronization of laser illumination, ultra-high-speed imaging, and ultrasound was accomplished using the same waveform generator.

Time-dependent, diffusion-driven growth of the ADV bubbles was acquired and analyzed using the manufacturer provided software (NIS-Elements AR 5.4, Nikon). ADV events were initiated approximately 1 mm above the cover slip and at x-y positions of at least 5R<sub>max</sub> away from any previously exposed location or edge of the hydrogel.

## 2.4. Quasistatic growth of ADV bubbles in fibrin-based hydrogels

Diffusion-driven ADV bubble growth recorded by confocal imaging ( $\sim O(s)$ ) was further fit to a modified Epstein-Plesset theory including the material's nonlinear elastic properties to calculate the growth rate. In the original Epstein-Plesset formula [43], the gas pressure in a bubble at equilibrium is given by  $p_g = p_\infty + \frac{2\gamma}{R}$ . When accounting for the elastic behavior of the material, the gas pressure can be calculated by adding a material stress term  $S$  corresponding to the elastic behavior of the material. The form can be chosen for a suitable material model in general, and for our choice of Neo-Hookean-like elasticity,  $S$  is given by [44]:  $S = \frac{G}{2} \left( 5 - 4 \left( \frac{R_0}{R} \right) - \left( \frac{R_0}{R} \right)^4 \right)$ . Following the approaches of [45,46] and incorporating the material stress term, we determined the relation for the quasistatic bubble radius growth rate in fibrin hydrogels to be

$$\frac{dR}{dt} = D \left( \frac{1}{R} + \frac{1}{\sqrt{\pi D t}} \right) \left( \frac{C_\infty - C_s \left( 1 + \frac{2\sigma}{R p_\infty} + S \right)}{\rho_g \left( 1 + \left( \frac{4\sigma}{3 R p_\infty} + S^* \right) \right)} \right), \quad (1)$$

where  $D$  is mass diffusion coefficient ( $D = 2 \times 10^{-9}$  m<sup>2</sup>/s for air in water),  $\rho_g$  is gas density in the bubble (1.2 kg/m<sup>3</sup> for air at ambient pressure),  $p_\infty$  is ambient pressure (101 kPa),  $G$  is the linear shear elastic modulus of fibrin hydrogels [41],  $C_\infty$  is gas concentration (kg/m<sup>3</sup>) away from the bubble,  $C_s$  is gas concentration at the bubble surface, and  $\gamma$  is surface tension (0.065 N/m) [47]. The constant  $c_s$  was determined based on Henry's law for oxygen and nitrogen gases in water at 37 °C at the ambient pressure and dimensionless ratio  $\frac{C_\infty - C_s}{\rho_g}$  was treated as a fitting parameter [45]. The effective elastic component,  $S^*$  for the Neo-Hookean elastic parameter was found to be

$$S^* = \frac{G}{6 p_\infty} \left( 15 - 8 \left( \frac{R_{sf}}{R} \right) + \left( \frac{R_{sf}}{R} \right)^4 \right), \quad (2)$$

where  $R_{sf}$  represents the equilibrium radius at which the surrounding medium is strain-free. The ratio of  $R_{sf}/R$  was varied as a fitting parameter to reach the best fit. Viscoelastic properties of fibrin-based hydrogels, under quasi-static conditions, were determined using a rheometer.

## 2.5. Statistics

Statistical analyses were performed in MATLAB using one-way ANOVA test. The data are displayed as the mean  $\pm$  standard deviation. We reported the number of independent replicates ( $n$ ) in the caption for each figure. Significant differences between different conditions studied here are represented in box plots in [Supplementary figures](#).

## 3. Results

### 3.1. Effect of fibrin concentration on ADV dynamics of PFP droplets

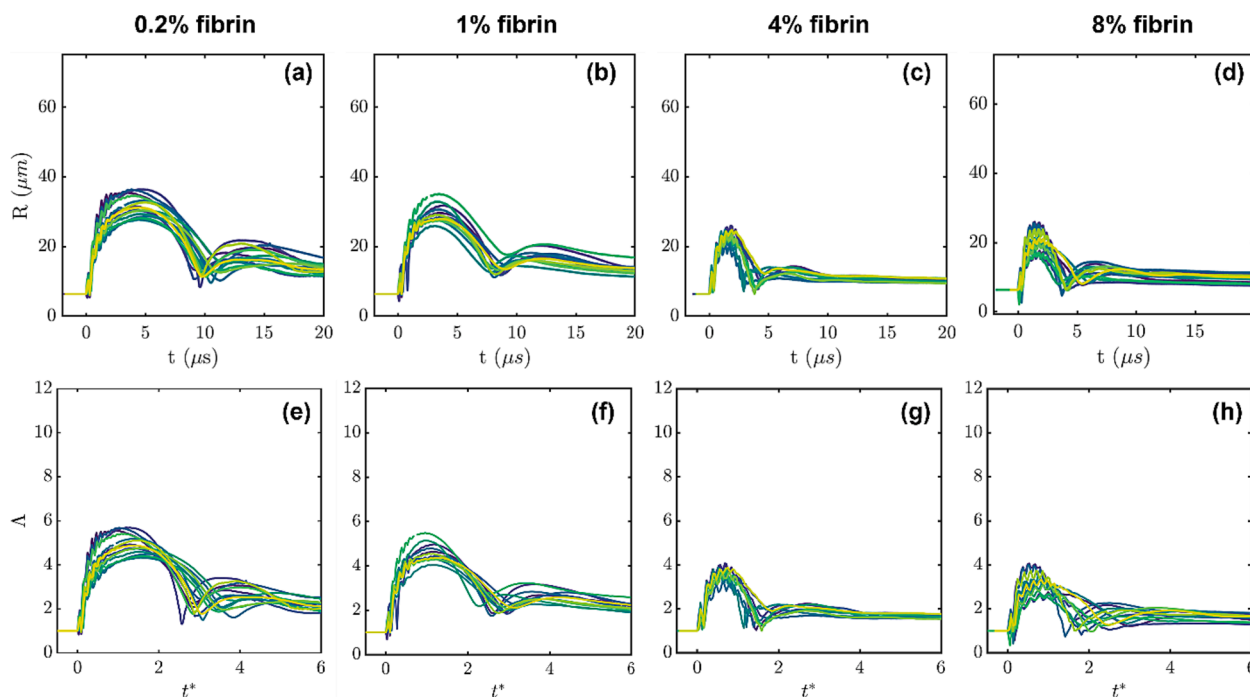
Higher fibrin concentration significantly increased both elastic and loss moduli in fibrin hydrogels as summarized in [Table 2](#). Spatiotemporally resolved ADV bubble dynamics, via ultra-high-speed imaging, are presented in [Supplementary videos 1a-d](#) for PFP droplets in fibrin hydrogels of varying concentrations under similar ultrasound parameters – a single burst of 2.4  $\mu$ s at 2.5 MHz ( $P_r$ : 4 MPa). At a frame rate of 10 Mfps, a time window of 20  $\mu$ s including before, during, and after ultrasound was recorded. Based on the change in the refractive index of a liquid droplet into a gas bubble, ultra-high-speed imaging indicated that phase-change was induced within  $0.8 \pm 0.2$   $\mu$ s following the arrival of ultrasound. Fibrin concentration did not significantly impact the initiation of ADV. ADV dynamics of PFP droplets, including expansion, subsequent oscillations, collapse, and rebounds are shown in bubble radius ( $R$ ) versus time ( $t$ ) plots (Fig. 2a-d) at varying fibrin concentrations. Each  $R$ - $t$  curve was further normalized by the initial radius of a PFP droplet ( $R_0$ ), termed the expansion ratio ( $\Lambda = R/R_0$ ), and the characteristic Rayleigh collapse time ( $t_{RC}$ ) ( $t^* = t/t_{RC}$ ), where  $t_{RC} = R_{max} \sqrt{\rho/p_\infty}$ ,  $p_\infty = 101.3$  kPa is the far-field pressure, and  $\rho = 1060$  kg/m<sup>3</sup> is the mass density of fibrin hydrogels [44,48] (Fig. 2e-h). From Fig. 2, key parameters of evolution of radial dynamics, as shown schematically in Fig. 1b, were extracted for each fibrin concentration and summarized in [Table 3](#). Data in [Table 3](#) were further analyzed for statistical significance for each condition and are plotted in [Fig. S1](#).

The key parameters are as follows: maximum bubble radius ( $R_{max}$ ) at the peak time ( $t = t_{peak}$ ), maximum expansion ratio ( $\Lambda_{max}$ ) defined as the ratio of an expanded bubble during ADV to the corresponding initial droplet radius ( $R_0$ ), the minimum bubble radius ( $R_c$ ) during collapse, collapse time ( $t_c$ ) which is the time when the bubble reached its maximum size to its final collapse radius, rebound radius ( $R_{reb}$ ), resting bubble radius at the end of high-speed microscopy at  $t = 20$   $\mu$ s ( $R_{rest}$ ), and maximum expansion velocity ( $V_{max}$ ).  $R_{max}$  correlated inversely with fibrin concentration, decreasing by  $\sim 45$  % when fibrin concentration increased from 0.2 % to 8 % (w/v). The effect of fibrin concentration was further reflected in  $\Lambda_{max}$ , showing significant differences ( $p < 0.001$ ) among all concentrations except for 0.2 % and 1 % fibrin hydrogels ([Fig. S1](#)). This observation has important implications for biomedical applications, since larger  $\Lambda_{max}$  values lead to larger strains and higher strain rates to the surrounding media. At  $t_{peak}$ , the surrounding fibrin hydrogel is out of dynamic equilibrium, with surface tension and material stress greater than the internal bubble pressure, ultimately driving the bubble collapse. At bubble collapse, the stored potential energy of the expanded bubble is converted into kinetic energy of the surrounding medium [49]. Increasing fibrin concentration reduced the initial collapse time, due to increased medium elasticity. During the collapse phase, the bubble contents compressed until the pressure inside the

**Table 2**

Bulk shear elastic and loss moduli of fibrin hydrogels were characterized using a rheometer.

| Fibrin concentration (%) | 0.2               | 1               | 4              | 8              |
|--------------------------|-------------------|-----------------|----------------|----------------|
| Elastic modulus (kPa)    | $0.18 \pm 0.01$   | $0.8 \pm 0.3$   | $6.5 \pm 1.0$  | $10.1 \pm 1.4$ |
| Loss modulus (kPa)       | $0.007 \pm 0.003$ | $0.04 \pm 0.01$ | $0.4 \pm 0.06$ | $1.0 \pm 0.1$  |



**Fig 2.** Acoustic droplet vaporization (ADV) dynamics of perfluoropentane (PFP) phase-shift droplets were recorded via ultra-high-speed microscopy at 10 Mfps. PFP droplets were embedded in hydrogels of varying fibrin concentrations (0.2–8 % w/v) and exposed to ultrasound at the following parameters: a single burst of 2.4  $\mu$ s and 2.5 MHz ( $P_r$ : 4 MPa). Bubble radius ( $R$ ) versus time ( $t$ ) curves (a–d) and the corresponding expansion ratio ( $\Lambda = R/R_0$ ) versus normalized time ( $t^*$ ) (non-dimensionalized by the Rayleigh collapse time ( $t_{RC}$ ) ( $t^* = t/t_{RC}$ )) (e–h) are shown for varying fibrin concentrations ( $n = 15$  for 0.2 % and 1 % fibrin,  $n = 12$  for 4 % fibrin, and  $n = 13$  for 8 % fibrin).

**Table 3**

Summary of parameters derived from ultra-high-speed and confocal microscopy studies (Figs. 2 & 3) for perfluoropentane phase-shift droplets embedded in fibrin-based hydrogels at varying fibrin concentrations. Statistical analyses of the parameters are shown in Fig. S1.

| Symbols         |            | Description           | Fibrin concentration (w/v) |                |                |                |
|-----------------|------------|-----------------------|----------------------------|----------------|----------------|----------------|
|                 |            |                       | 0.2 %                      | 1 %            | 4 %            | 8 %            |
| $R_{max}$       | ( $\mu$ m) | maximum radius        | $31.9 \pm 2.9$             | $29.7 \pm 2.6$ | $23.5 \pm 1.5$ | $18.0 \pm 2.1$ |
| $t_{peak}$      | ( $\mu$ s) | time to 1st peak      | $6.0 \pm 0.5$              | $5.8 \pm 0.3$  | $2.7 \pm 0.3$  | $2.2 \pm 0.4$  |
| $\Lambda_{max}$ |            | maximum expansion     | $5.3 \pm 0.5$              | $4.9 \pm 0.5$  | $3.8 \pm 0.2$  | $3.0 \pm 0.5$  |
| $R_c$           | ( $\mu$ m) | collapse radius       | $12.0 \pm 2.2$             | $12.2 \pm 1.9$ | $8.5 \pm 1.9$  | $7.4 \pm 1.3$  |
| $t_c$           | ( $\mu$ s) | collapse time         | $5.9 \pm 0.8$              | $5.6 \pm 0.4$  | $2.7 \pm 0.4$  | $2.1 \pm 0.5$  |
| $R_{reb}$       | ( $\mu$ m) | radius at 1st rebound | $17.6 \pm 2.3$             | $17.4 \pm 1.7$ | $13.3 \pm 1.0$ | $12.0 \pm 1.6$ |
| $V_{max}$       | (m/s)      | expansion velocity    | $48.7 \pm 8.5$             | $53.1 \pm 7.3$ | $48.3 \pm 7.4$ | $45.3 \pm 9.0$ |
| $R_{rest}$      | ( $\mu$ m) | radius at 20 $\mu$ s  | $13.1 \pm 1.5$             | $13.2 \pm 1.3$ | $10.1 \pm 0.5$ | $9.2 \pm 1.0$  |
| $R_{eq}$        | ( $\mu$ m) | radius at 60 s        | $52.2 \pm 2.6$             | $43.0 \pm 1.9$ | $35.4 \pm 2.6$ | $32.5 \pm 3.1$ |

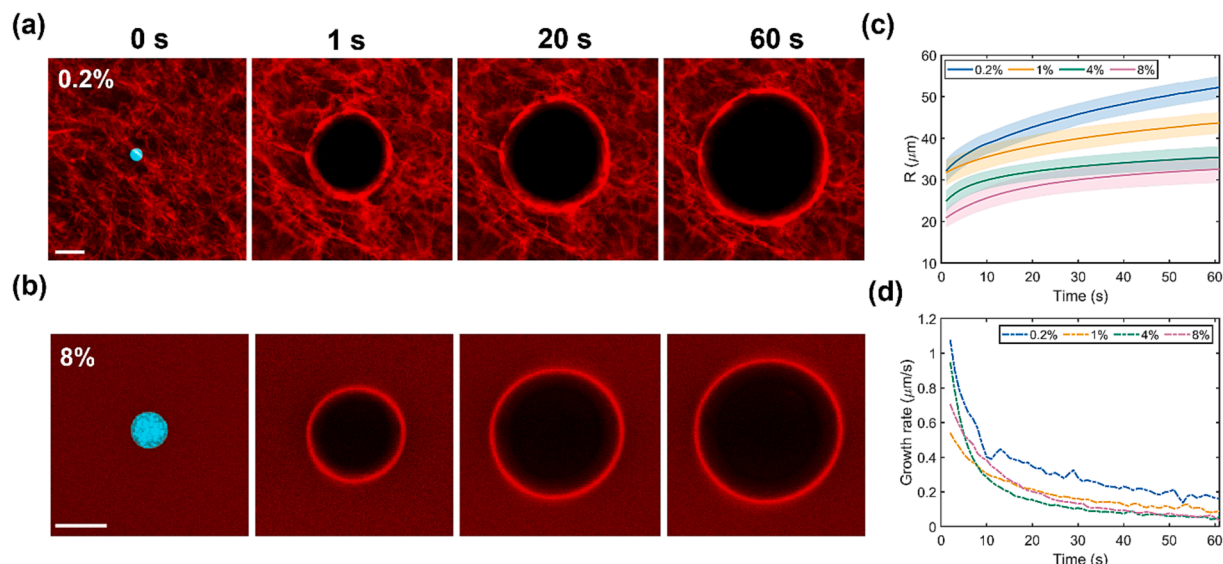
bubble was sufficiently large to balance the contraction forces [50]. The collapse time,  $t_c$ , was significantly shorter in higher fibrin concentrations. The collapse was completely arrested at  $12.1 \pm 0.8 \mu$ s,  $11.0 \pm 0.6 \mu$ s,  $5.1 \pm 0.6 \mu$ s, and  $4.9 \pm 0.8 \mu$ s in hydrogels of 0.2 %, 1 %, 4 %, and 8 % fibrin concentrations, respectively, following which the bubble began to rebound.  $R_{reb}$  was significantly smaller than  $R_{max}$  in all fibrin concentrations, most likely due to viscous dissipation and acoustic radiation during the initial collapse [51]. The amplitude of  $R_{reb}$ , indicative of the effect of fibrin viscosity, inversely correlated with fibrin concentration. However, the ratio of  $R_{reb}/R_{max}$  was  $0.5 \pm 0.05$ , which was not statistically significant among the four fibrin concentrations studied.

There was no statistically significant difference in  $V_{max}$  among the different fibrin concentrations (Table 3, Fig. S1). Post-ultrasound dynamics was also impacted by fibrin concentration.  $R_{rest}$  of the resultant ADV bubbles from PFP droplets was significantly larger in 0.2 % (w/v) fibrin hydrogels compared to 4 % (w/v) (p-value < 0.0001) and 8 % (w/v) (p-value < 0.0001) fibrin hydrogels. The diffusion-driven growth of ADV bubbles, because of gas influx, was further captured by confocal imaging up to 60 s (Fig. 3).

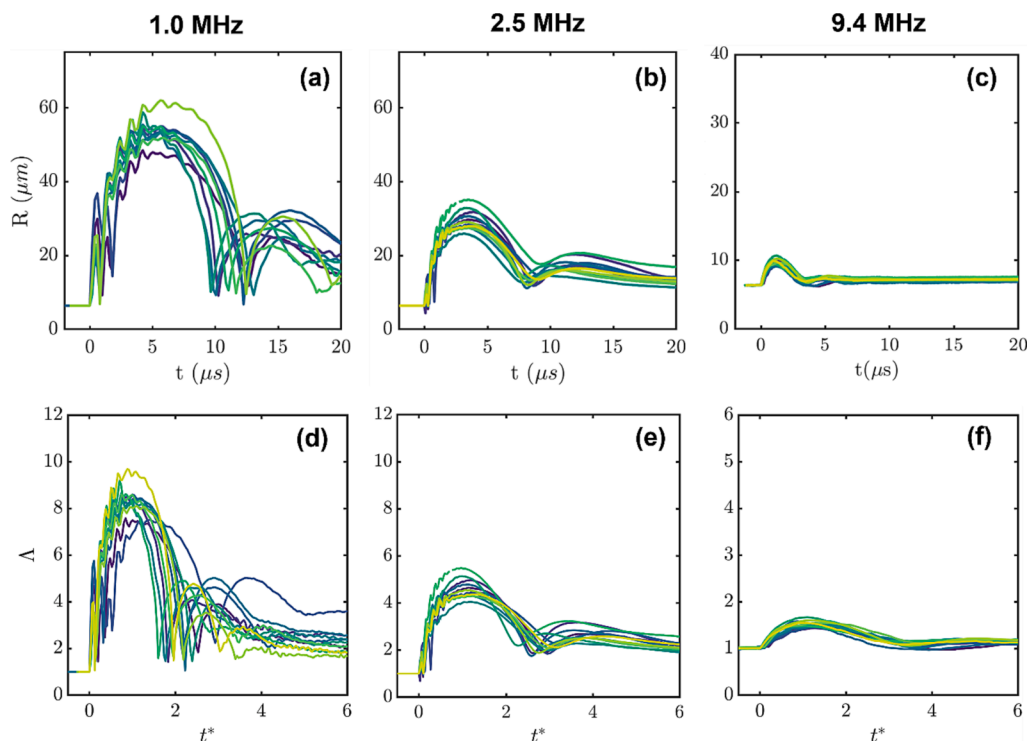
As shown in Fig. 3, growth of the bubbles was hindered in higher fibrin concentrations. The time scale of passive diffusion ( $\sim O(s)$ ) was six orders of magnitude slower than the ADV dynamics ( $\sim O(\mu s)$ ).  $R_{eq}$  was significantly different among all four fibrin concentrations (p-value < 0.001), being larger in lower fibrin concentrations (Fig. S1). The modified Epstein-Plesset equation (Eq. (1)), accounting for surface tension and fibrin elasticity, was used to estimate diffusion-driven growth rate of the ADV-bubbles. Differences in growth rates of ADV-induced bubbles in fibrin-based hydrogels are shown in Fig. 3d.

### 3.2. Effect of frequency of excitation on the ADV dynamics of PFP droplets

A single burst consisting of 6 cycles at 4 MPa peak rarefactional pressure generated ADV at three clinically relevant frequencies. The effect of excitation frequency (1 MHz, 2.5 MHz, and 9.4 MHz) on the ADV dynamics of PFP droplets embedded in 1 % (w/v) fibrin hydrogels is presented in Supplementary videos 5–7. The corresponding R-t plots are shown in Fig. 4. Initiation of ADV, based on change in the refractive



**Fig 3.** Longitudinal, diffusion-driven growth of perfluoropentane (PFP) phase-shift droplets vaporized via acoustic droplet vaporization (ADV) was recorded via confocal microscopy. PFP droplets (shown in cyan) were embedded in varying fibrin concentrations (0.2–8 % w/v) (shown in red). Selected confocal images of growing PFP bubbles in (a) 0.2 % and (b) 8 % fibrin hydrogels are shown. (c) Longitudinal growth of ADV bubbles up to 60 s is plotted for each fibrin concentration. (d) Modified Epstein-Plesset equation (Eq. (1)) was used to calculate the growth rate of ADV bubbles in each fibrin concentration. Acoustic parameters were as follows: a single burst of 2.4  $\mu$ s at 2.5 MHz ( $P_r$ : 4 MPa). Scale bar: 25  $\mu$ m.



**Fig 4.** Acoustic droplet vaporization (ADV) dynamics of perfluoropentane (PFP) phase-shift droplets at three clinically relevant frequencies were recorded via ultra-high-speed microscopy at 10 Mfps. PFP droplets embedded in 1 % fibrin hydrogels. Bubble radius ( $R$ ) versus time ( $t$ ) curves (a-c) and the corresponding expansion ratio ( $\Lambda = R/R_0$ ) versus normalized time ( $t^*$ ) (non-dimensionalized by the Rayleigh collapse time ( $t_{RC}$ )) ( $t^* = t/t_{RC}$ ) (d-f) are shown for three excitation frequencies of 1 MHz, 2.5 MHz, and 9.4 MHz. A single burst consisting of 6 cycles at 4 MPa peak rarefactional pressure was used to generate ADV ( $n = 11$  for 1 MHz,  $n = 12$  for 2.5 MHz, and  $n = 13$  for 9.4 MHz).

index from liquid to gas, was significantly shorter at 9.4 MHz compared to the other two frequencies (p-value = 0.002). Parameters extracted from  $R$ - $t$  (Fig. 4a-c) and the corresponding  $\Lambda$ - $t^*$  curves (Fig. 4d-f) were further analyzed for statistical significance (Fig. S2), indicating a significant impact of the excitation frequency on the radial dynamics of PFP

droplets.

ADV-induced PFP bubbles had a significantly higher  $\Lambda_{\max}$  at 1 MHz compared to 2.5 MHz (p-value < 0.0001) and 9.4 MHz (p-value < 0.0001). As can be observed in [Supplementary videos 5–7](#) and [Fig. 4](#), ADV was more localized and confined within PFP phase-shift droplets at

**Table 4**

Summary of parameters derived from ultra-high-speed and confocal microscopy studies (Figs. 4 & 5) for perfluoropentane phase-shift droplets embedded in 1% (w/v) fibrin-based hydrogels and exposed to ultrasound at three different excitation frequencies. Definition of parameters is given in Table 3 and shown schematically in Fig. 1b.

| Symbols           |                   | Frequency (MHz) |                |                 |
|-------------------|-------------------|-----------------|----------------|-----------------|
|                   |                   | 1               | 2.5            | 9.4             |
| $R_{\max}$        | ( $\mu\text{m}$ ) | $55.5 \pm 3.6$  | $29.7 \pm 2.6$ | $7.6 \pm 0.5$   |
| $t_{\text{peak}}$ | ( $\mu\text{s}$ ) | $7.4 \pm 1.3$   | $5.8 \pm 0.3$  | $2.4 \pm 0.2$   |
| $\Lambda_{\max}$  |                   | $9.1 \pm 0.6$   | $4.9 \pm 0.5$  | $1.3 \pm 0.1$   |
| $R_c$             | ( $\mu\text{m}$ ) | $9.7 \pm 1.3$   | $12.2 \pm 1.9$ | $3.3 \pm 0.6$   |
| $t_c$             | ( $\mu\text{s}$ ) | $6.2 \pm 0.7$   | $5.1 \pm 0.5$  | $2.0 \pm 0.2$   |
| $R_{\text{reb}}$  | ( $\mu\text{m}$ ) | $28.1 \pm 3.2$  | $17.4 \pm 1.7$ | $4.7 \pm 0.4$   |
| $V_{\max}$        | (m/s)             | $58.4 \pm 17.3$ | $54.1 \pm 7.3$ | $23.4 \pm 13.5$ |
| $R_{\text{rest}}$ | ( $\mu\text{m}$ ) | $12.8 \pm 1.5$  | $13.2 \pm 1.3$ | $9.4 \pm 0.8$   |
| $R_{\text{eq}}$   | ( $\mu\text{m}$ ) | $43.2 \pm 2.6$  | $43.0 \pm 1.9$ | $33.7 \pm 1.6$  |

9.4 MHz. Additionally,  $t_c$  significantly reduced, up to two-fold, as the excitation frequency increased. Similar to trends observed with fibrin concentration, higher excitation frequency significantly reduced the amplitude of  $R_{\text{reb}}$  (Table 4, Fig. S2). Although  $V_{\max}$  was not significantly different at the two excitation frequencies of 1 MHz, and 2.5 MHz, it decreased by up to 60 % at an excitation frequency of 9.4 MHz (Fig. S2).

Stable bubble formation was observed at all excitation frequencies, however the resultant ADV bubbles had significantly smaller  $R_{\text{rest}}$  and  $R_{\text{eq}}$  at 9.4 MHz compared to 2.5 MHz and 1 MHz (Fig. 5). Differences in growth rates of ADV-induced bubbles at different frequencies are shown in Fig. 5d. Table 4 summarizes the derived parameters from the experimental high-speed and confocal microscopy studies at varying frequencies.

### 3.3. Effect of bulk boiling point of phase-shift droplets on the ADV dynamics

In addition to the surrounding media viscoelasticity and excitation frequency, bulk boiling point of phase-shift droplets is another important determinant of phase-change and cavitation properties. Representative ultra-high-speed ADV dynamics for PFP, PFH and PFO droplets in 1 % (w/v) fibrin hydrogels are shown in Supplementary videos 6–8 and plotted in Fig. 6. ADV initiation was not significantly different among

the three phase-shift droplets and occurred within a microsecond after the arrival of ultrasound.

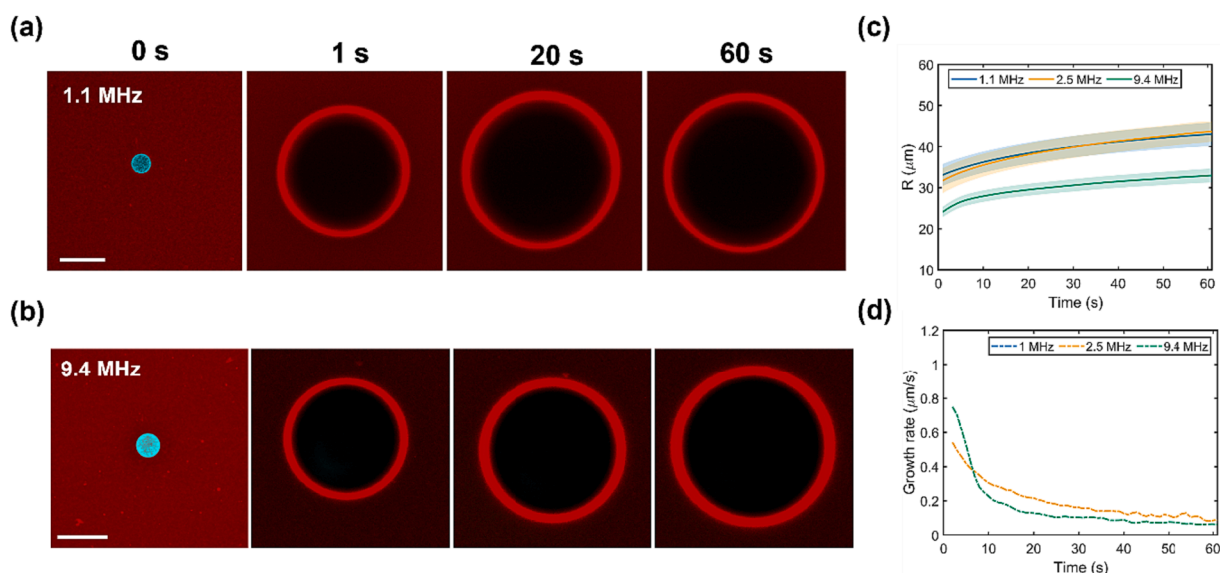
Bulk boiling point of phase-shift droplets significantly affected parameters during and post ultrasound (Table 5, Fig. S3). For example,  $R_c$ ,  $R_{\text{rest}}$ ,  $R_{\text{eq}}$  were significantly smaller in PFH and PFO droplets compared to PFP droplets, most likely due to a significantly higher vapor pressure and as a result higher internal pressure inside the generated PFP bubble.

Although the ratio of  $R_{\text{rest}} / R_0$  was  $\sim 1.1$  for both PFH and PFO droplets, indicating partial or complete recondensation, confocal microscopy consistently captured stable bubble formation and micropore formation for PFH and PFO droplet, respectively, on a longer timescale ( $\sim O(s)$ ) (Fig. 7). Due to a significantly lower vapor pressure, no stable bubble formation was observed for PFO droplets after ultrasound was turned off, evident by an insignificant change in temporal evolution of radius in the recorded confocal images (Fig. 7c & d).

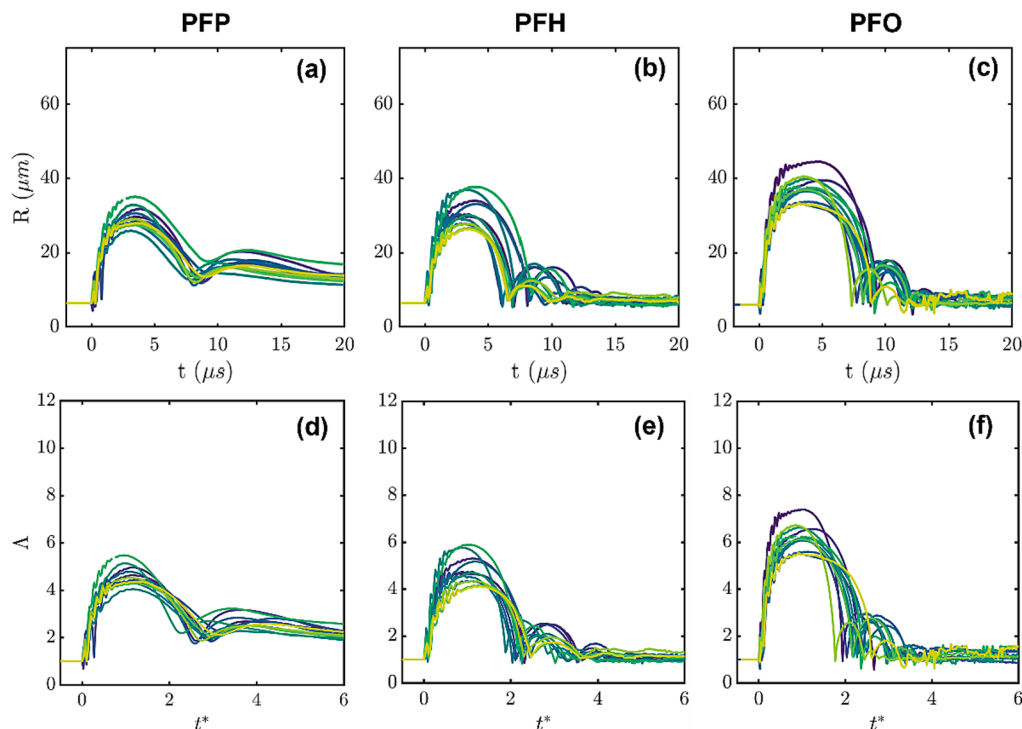
## 4. Discussion

We integrated ultra-high-speed imaging, confocal microscopy, and focused ultrasound for full-scale characterization of ADV dynamics of three commonly used phase-shift droplets within a viscoelastic environment. Radial dynamics of ADV bubbles, key to their effectiveness for both imaging and therapeutic applications, are a function of intrinsic features (e.g., droplet size and bulk boiling point) and extrinsic factors (e.g., driving frequency and surrounding media). Given the potential applications of phase-shift droplets in biomedical applications, there is a need to extend our understanding to viscoelastic media. The effect of surrounding viscoelasticity has not been explored in the past since prior experimental high-speed studies of ADV dynamics were conducted primarily in Newtonian fluids [21–23,52]. Fibrin was chosen as the hydrogel for its broad range of biomedical applications [27,28]. Due to the high temporal resolution of our ultra-high-speed camera (up to 10 Mfps), different stages from liquid to gas expansion, oscillation of the generated ADV bubbles, to their collapse and subsequent rebound were captured, as can be seen in the R-t plots presented in the manuscript.

Our rheological measurements, under quasistatic conditions, showed that both elastic (storage) and loss moduli correlated with fibrin concentration (Table 2). Understanding how changes in the microstructure and stiffness of the surrounding media impact radial dynamics of ADV bubbles is critical since the mechanical properties of tissues vary and can



**Fig 5.** Confocal microscopy was used to record the longitudinal growth of bubbles generated via acoustic droplet vaporization at different frequencies. PFP droplets (shown in cyan) were embedded in 1 % (w/v) fibrin hydrogels (shown in red) and exposed to a single burst ultrasound (6 cycles) at (a) 1.1 MHz ( $P_r$ : 4 MPa), and (b) 9.4 MHz ( $P_r$ : 4 MPa). (c) Time-dependent growth of ADV bubbles up to 60 s are shown. (d) Modified Epstein-Plesset equation (Eq. (1)) was used to calculate the growth rate of ADV bubbles in 1 % fibrin at varying frequencies. Scale bar: 25  $\mu\text{m}$ .



**Fig 6.** Acoustic droplets vaporization (ADV) dynamics of phase-shift droplets with different bulk boiling points were recorded via ultra-high-speed microscopy. Three perfluorocarbon liquid cores were used: perfluoropentane (PFP), perfluorohexane (PFH), and perfluorooctane (PFO). Bubble radius ( $R$ ) versus time ( $t$ ) curves (a-c) and the corresponding expansion ratio ( $\Lambda = R/R_0$ ) versus normalized time ( $t^*$ ) (non-dimensionalized by the Rayleigh collapse time ( $t_{RC}$ ) ( $t^* = t/t_{RC}$ )) (d-f) are shown for three perfluorocarbon phase-shift droplets. Acoustic parameters were as follows: a single burst of  $2.4 \mu\text{s}$  at  $2.5 \text{ MHz}$  ( $P_r$ :  $4 \text{ MPa}$ ). A higher pressure ( $P_r$ :  $4.5 \text{ MPa}$ ) was used for PFO droplets ( $n = 12$  for each phase-shift droplet).

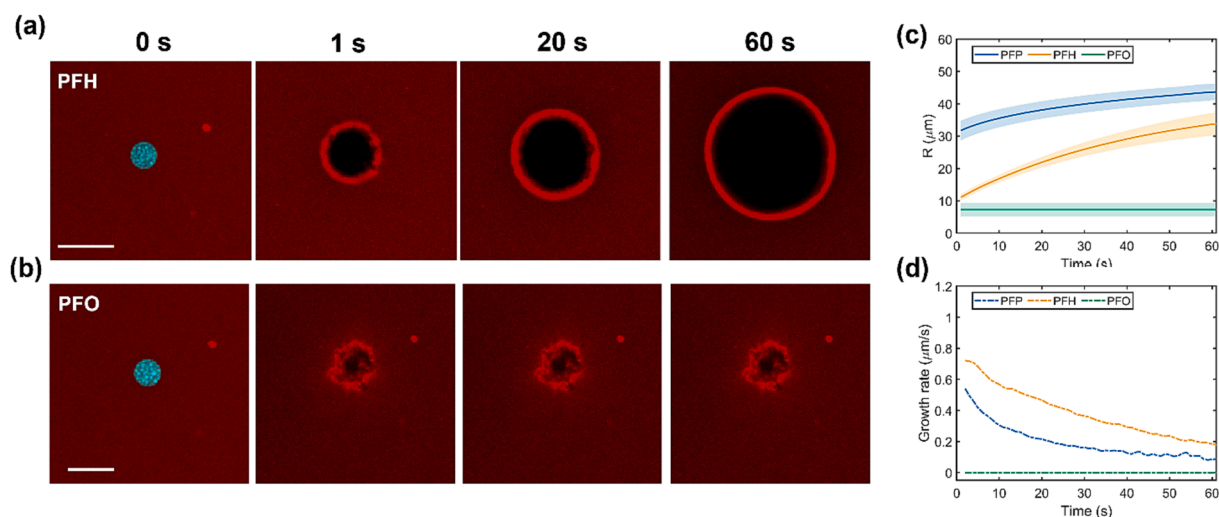
**Table 5**

Summary of parameters derived from ultra-high-speed and confocal microscopy studies (Figs. 6 & 7) for three phase-shift droplets embedded in 1% (w/v) fibrin-based hydrogels: perfluoropentane (PFP), perfluorohexane (PFH), and perfluorooctane (PFO). Definition of parameters is given in Table 3 and schematically shown in Fig. 1b.

| Symbols                |                   | Phase-shift droplet |                |                 |
|------------------------|-------------------|---------------------|----------------|-----------------|
|                        |                   | PFP                 | PFH            | PFO             |
| $R_{\text{max}}$       | ( $\mu\text{m}$ ) | $29.7 \pm 2.6$      | $30.9 \pm 3.8$ | $37.4 \pm 3.5$  |
| $t_{\text{peak}}$      | ( $\mu\text{s}$ ) | $5.8 \pm 0.3$       | $4.8 \pm 0.5$  | $5.4 \pm 0.5$   |
| $\Lambda_{\text{max}}$ |                   | $4.9 \pm 0.5$       | $5.1 \pm 0.6$  | $5.7 \pm 0.6$   |
| $R_c$                  | ( $\mu\text{m}$ ) | $12.2 \pm 1.9$      | $6.7 \pm 1.1$  | $6.9 \pm 1.4$   |
| $t_c$                  | ( $\mu\text{s}$ ) | $5.1 \pm 0.5$       | $3.6 \pm 0.5$  | $4.5 \pm 0.4$   |
| $R_{\text{reb}}$       | ( $\mu\text{m}$ ) | $17.4 \pm 1.7$      | $14.0 \pm 2.0$ | $16.7 \pm 2.4$  |
| $V_{\text{max}}$       | ( $\text{m/s}$ )  | $53.1 \pm 7.3$      | $49.5 \pm 7.5$ | $68.0 \pm 14.0$ |
| $R_{\text{rest}}$      | ( $\mu\text{m}$ ) | $13.2 \pm 1.3$      | $7.0 \pm 0.7$  | $7.6 \pm 1.2$   |
| $R_{\text{eq}}$        | ( $\mu\text{m}$ ) | $43.0 \pm 1.9$      | $33.7 \pm 3.5$ | $7.6 \pm 2.2$   |

change as a result of disease. For example, elastic moduli in healthy and cancer pancreatic tissues were reported to be around  $1 \text{ kPa}$  and  $6 \text{ kPa}$ , respectively. In the current study, ADV bubbles achieved a larger maximum radius in lower fibrin concentrations. As fibrin concentration increased from  $0.2 \text{ % (w/v)}$  to  $8 \text{ % (w/v)}$ ,  $\Lambda_{\text{max}}$  significantly decreased by  $\sim 50 \text{ %}$  (Fig. 2 and Table 3).  $\Lambda_{\text{max}}$  is an important quantity for developing safe and effective diagnostic and therapeutic applications, as our prior work has found correlation between maximum strain at bubble walls with damage in cell cultures exposed to inertial cavitation [53]. Larger  $\Lambda_{\text{max}}$  in ADV-induced bubbles similarly leads to larger strains to the surrounding. Ultra-fast tracking of particle displacements induced by ADV indicated that radial strains (engineering strains) were largest at maximum bubble radius and correlated directly with  $\Lambda_{\text{max}}$  [54]. A radial strain of  $\sim 2$  was measured for a PFP bubble with a  $\Lambda_{\text{max}}$  of  $\sim 3.5$  in  $1 \text{ %}$  fibrin hydrogels. The ADV-induced strains were hyperlocal and

decreased significantly with distance from the bubbles [54]. Radial strain fields (logarithmic Hencky strain) were shown to decay as  $\sim 1/r^3$ , where  $r$  is the distance from the center of the laser-induced bubble [55]. In earlier study of laser-induced bubbles in polyacrylamide hydrogels, using the quadratic law Kelvin-Voigt model to represent soft materials with strain-stiffening behavior, elastic stresses were shown to be highest at  $R_{\text{max}}$ , where strain was largest, while peaks in viscous stress occurred at the onset of bubble growth and at  $R_c$ , where strain rate was largest [56]. In the context of mechanical characterization of soft biomaterials at high strain rates, dynamics of laser-induced bubbles in gelatin [57] and agarose [58] hydrogels were studied via ultra-high-speed imaging. Similar to our observation, a 10-fold increase in agarose concentration reduced  $R_{\text{max}}$  of laser-induced bubbles ( $R_{\text{max}}$ :  $\sim 260 \mu\text{m}$  in  $0.5 \text{ %}$  agarose vs  $R_{\text{max}}$ :  $\sim 184 \mu\text{m}$  in  $5 \text{ %}$  agarose). This emphasizes the sensitivity of bubble dynamics to surrounding properties as previously supported by several numerical studies [26,44,59]. Specifically, an increase of fibrin concentration increases elastic modulus, which serves to decrease the maximum attainable radius of the bubble via stress built up in the surrounding material during expansion. Notably, the sensitivity to the surrounding viscoelasticity was shown numerically for micron-sized bubbles with radii smaller than  $50 \mu\text{m}$  [60]. Changes in radial dynamics as a result of the surrounding media can significantly impact the scattering cross section of the generated bubbles and thus their acoustic emissions, both linear resonance properties and nonlinear behavior, making ADV bubbles suitable probes of local tissue characterization. As fibrin concentration increased, growth and collapse velocities of the generated ADV bubbles were significantly faster (Fig. S1). Prior numerical investigations also showed that fluid elasticity accelerated the collapse of the bubbles [61,62]. According to Rayleigh collapse, larger  $R_{\text{max}}$  leads to a longer collapse time as was observed for PFP bubbles in lower fibrin concentration. ADV-generated bubbles settled to diameters significantly smaller than the maximum diameters reached during the expansion phase (i.e.,  $R_{\text{rest}} < R_{\text{max}}$ ). The ratio of  $R_{\text{max}}/R_{\text{rest}}$  was  $2.4$ ,  $2.2$ ,



**Fig 7.** Stable bubble formation via acoustic droplet vaporization depended on the bulk boiling point of phase-shift droplets. While stable bubbles were consistently generated in perfluoropentane (PFP) and perfluorohexane (PFH) phase-shift droplets embedded in hydrogels with 1 % (w/v) fibrin concentration, micropores were generated when perfluorooctane (PFO) droplets were used. (a) Representative confocal images of a PFH phase-shift droplet (shown in cyan) embedded in a fibrin hydrogel (shown in red) display diffusion-driven growth of the generated bubble up to 60 s after ultrasound. (b) ADV generated a fluid-filled micropore with no significant change in size over time for a PFO droplet. (c) Longitudinal growth of PFP, PFH, and PFO droplets are plotted from 1 s up to 60 s after ultrasound. (d) Modified Epstein-Plesset equation (Eq.1) was used to calculate the growth rate of the generated ADV features in 1 % fibrin. Droplets were exposed to a single burst of ultrasound (number of cycles: 6) at 2.5 MHz ( $P_r$ : 4 MPa) for PFP and PFH droplets. A higher pressure ( $P_r$ : 4.5 MPa) was used for PFO droplets. Scale bar: 25 μm.

2.3, and 1.9 for ADV-generated bubbles at 2.5 MHz in fibrin hydrogels of 0.2 % (w/v), 1 % (w/v), 4 % (w/v), and 8 % (w/v) with PFP phase-shift droplets, respectively. Knowing  $R_{rest}$  and  $R_{max}$  can help in applications such as sonoporation where the distance of cells with respect to phase-shift droplets (d) can define cellular bioeffects such as irreversible sonoporation ( $d < R_{rest}$ ), reversible sonoporation ( $R_{rest} < d < R_{max}$ ), or no sonoporation ( $d > R_{max}$ ) [54,63,64].

Fibrin concentration impacted temporal growth of the ADV generated bubbles in PFP droplets on a longer time scale as well. Time-lapse confocal imaging recorded the passive diffusion-driven growth (i.e., in the absence of a changing pressure field) of the ADV-induced bubbles up to 60 s after ultrasound was turned off. The slow growth rate of ADV-generated bubbles, which correlated inversely with fibrin concentration, may be driven by the layer of highly dense fibrin surrounding the bubble that provides shell-induced resistance to gas transfer [65]. The high driving pressure used here (i.e.,  $P_r$ : 4 MPa) led to non-spherical oscillations of ADV bubbles particularly in lower fibrin concentrations (Videos S1-4). However, the generated ADV-bubbles appeared spherical in confocal images, taken on a longer time scale compared to the ADV time scale, due to surface tension and stabilizing effects of the material's viscoelasticity at longer timescales. It should be noted that the ultrasound forcing period used here was on the order of 106 ns (at 9.4 MHz) to 1 μs (at 1 MHz), which was much shorter than the fibrin relaxation time ( $\sim O(ms)$ ) [66]. This led to strain buildup in the matrix during ADV. Since fibrin, like other natural biomaterials, exhibits strain stiffening behavior, ADV-generated bubbles can induce significant compaction and stiffening at the bubble-fibrin interface [66,67]. ADV-induced strain stiffening, evident by significantly brighter fluorescence intensities in confocal images (Figs. 3, 5, and 7) post-ultrasound, was observed for all the three phase-shift droplets regardless of stable or transient bubble formation. This is important since ADV-induced stiffening in natural biomaterials can enable on-demand modulation (i.e., both spatially and temporally) of cellular phenotype and signaling [67,68].

Excitation frequency and pressure are two critical parameters associated with bubble dynamics. In this work, all three frequencies of ultrasound triggered ADV in PFP droplets. Frequency of excitation had the most pronounced effect on ADV dynamics of PFP droplets at a constant fibrin concentration.  $\Lambda_{max}$  at 1 MHz was 2 times higher than 2.5 MHz, and 7 times higher than 9.4 MHz (Table 4). This is expected since bubble

growth is inversely proportional to the excitation frequency since lower frequency ultrasound provides a longer window for growth of the nucleated bubble via rectified diffusion [69]. A longer rarefactional pressure at low excitation frequency results in a larger maximum bubble diameter. In applications where maintaining an upper limit on bubble size is critical, careful consideration should be given to the selection of the frequency. Higher excitation frequencies also enable better spatial selectivity of ADV due to a smaller focal volume as well as lower probability of inertial cavitation related damage to large molecular payloads or adjacent cells. In our studies, initiation of ADV was faster at 9.4 MHz likely due to superharmonic focusing being more prominent in micron-size droplets (larger than 6 μm in diameter) at higher frequencies where the wavelength of higher harmonics becomes comparable to the droplet size [22,23].

Ultra-fast dynamics of phase-shift droplets with higher bulk boiling points remain less explored. Despite the vapor pressure in PFP droplets being three times higher than PFH droplets at the ambient condition (Table 1), there was no significant difference in  $\Lambda_{max}$ . This is likely due to the vapor pressure ( $P_v$ ) becoming less significant as the driving acoustic pressure increases (i.e.,  $P_r \gg P_v$ ). Using classical nucleation theory, we demonstrated that the dependence of the critical radius of nucleation on the saturated vapor pressure of perfluorocarbon liquids was greatly reduced at high peak rarefactional pressures [42]. However, post-ADV dynamics were distinct for each phase-shift droplet. Stable bubble formation in PFH droplets can be attributed to inward gas diffusion during ADV enhanced by the higher amplitude and long pulse durations used here (i.e., rectified diffusion). Contrastingly, PFO droplets recondensed. The notably reduced vapor pressure along with lower solubility and mass diffusivity of oxygen decreased the volume of non-condensed gas and, thus, the likelihood of survival of a bubble in a PFO droplet. Higher bulk boiling point phase-shift droplets such as PFO not only offer enhanced thermal stability (i.e., preventing spontaneous bubble formation), they also enable a unique potential for transient bubble formation (i.e., during ultrasound). PFO droplets consistently underwent fragmentation (Video S10) after ultrasound was turned off.

Future studies will explore the dynamics of phase-shift droplets when administered in populations since biomedical applications rarely rely on activation of a single droplet. Furthermore, we will investigate how the various ADV bubble dynamics observed in this study impact the release

kinetics of payload-carrying phase-shift droplets and the resulting bio-effects via ultra-high-speed fluorescence microscopy.

## 5. Conclusions

ADV provides a dynamic platform for on-demand generation of microbubbles for both diagnostic and therapeutic applications of ultrasound. We investigated the dynamics of ADV by combining ultra-fast and confocal microscopy techniques. The findings demonstrate that dynamics of ADV can be tuned by intrinsic features, such as bulk boiling point of phase-shift droplets, or extrinsic factors, such as surrounding media viscoelasticity and acoustic parameters. **ADV bubbles achieved a larger maximum radius in lower fibrin concentrations.** The significant impact of fibrin concentration on key parameters such as  $\Lambda_{\max}$  can open doors to a novel ADV-assisted tissue characterization technique. **Three clinically relevant frequencies triggered ADV in PFP droplets resulting in stable bubble formation,** although the dynamics were significantly different among the three frequencies during and post ultrasound. Similar to higher fibrin concentrations, **higher driving frequencies suppressed maximum bubble expansion radii.** We showed that under similar acoustic conditions, **stable or transient bubble formation depended on the bulk boiling point of the phase-shift droplets,** enabling unique ADV dynamics which can be tailored for specific applications. Overall, this work provides a full-scale characterization, from nano-second to seconds, of ADV-bubble dynamics under varying conditions which could aid in optimizing current applications as well as exploiting future opportunities of ADV.

## CRedit authorship contribution statement

**Bachir A. Abeid:** Formal analysis, Methodology, Visualization, Writing – original draft. **Mario L. Fabiilli:** Conceptualization, Methodology, Resources, Supervision, Writing – review & editing. **Jonathan B. Estrada:** Conceptualization, Funding acquisition, Resources, Writing – review & editing. **Mitra Aliabouzar:** Conceptualization, Investigation, Methodology, Resources, Supervision, Visualization, Writing – original draft, Writing – review & editing.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

Data will be made available on request.

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## Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ultsonch.2024.106754>.

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